

# CMPO v2.0 — Chemical-Mechanical Planarization Process Ontology

## Internal Specification

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## 1 Purpose and Scope

CMPO (Chemical-Mechanical Planarization Process Ontology) is a domain ontology for representing CMP process data — wafers, tools, tool components, consumables, process steps, recipes, and time-stamped sensor observations — as an RDF knowledge graph. Version 2.0 is the frozen reference version used by two Master's theses at the professorship (semantic transformation/validation/queriing, and LLM-based natural-language queriing) and by the associated research on AI-assisted ontology construction. The empirical anchor is the PHM 2016 Data Challenge dataset (CMP wafer polishing); the ontology is designed to generalize to comparable industrial time-series settings.

Files: `cmpos-v2.0.ttl` (core, this specification) and `cmpos-bfo-alignment-v2.ttl` (optional alignment module, Section 6). Turtle syntax; namespace:

```
@prefix cmpos: <https://vsr.informatik.tu-chemnitz.de/ontologies/cmpos#>
```

## 2 Design Decisions

- **SOSA/SSN backbone.** The W3C sensor-observation standard provides the core pattern: every measured value is one `sosa:Observation`. Rationale: PHM data IS sensor observations; the pattern gives every CSV cell one canonical home, one canonical SPARQL access path, and standards-level interoperability and publishability.
- **Provenance and construction method.** CMPO was constructed AI-assisted from domain-expert input and refactored onto the SOSA/SSN backbone; v2.0 supersedes the unversioned working schema ("Schema\_v2"), resolving its structural conflicts (multi-domain properties, dual-parent classes, class/property ambiguities) and fixing its namespace.
- **BFO kept separate.** Top-level (BFO) anchors live in an optional alignment module, not in the core — application graphs stay SOSA-only, while the module enables comparison with BFO-based ontologies.
- **Naming.** UpperCamelCase classes, lowerCamelCase properties. Source CSV column names are preserved as `skos:altLabel` on observable properties (e.g. "RETAINER\_RING\_PRESSURE") for exact data-to-schema grounding and NL retrieval. The identifier/context headers (MACHINE\_ID, TIMESTAMP, WAFER\_ID, STAGE, CHAMBER) likewise carry `skos:altLabel` on their datatype properties (`toolId`, `rawTimestamp`, `hasWaferId`, `stageLabel`, `chamberNumber`), so all 28 PHM headers ground 1:1 via `altLabel`.
- **Documentation completeness.** Every class and property carries `rdfs:label` and `rdfs:comment` (100% coverage; 165 classes, 116 properties (53 object, 63 datatype), ~1,380 triples). Comments double as the retrieval corpus for schema-aware LLM prompting.
- **Units.** Unit individuals aligned to QUDT via `owl:sameAs`; observable properties declare a `cmpos:canonicalUnit`. Parameter classes are additionally used in individual positions (as `sosa:observedProperty` targets and `cmpos:canonicalUnit` subjects) via OWL 2 class-individual punning; this is intentional and documented in the ontology header.

## 3 The Observation Pattern (Core)

One CSV cell = one observation:

```
:obs_00001 a cmpos:CMPObservation ;
  sosa:observedProperty      cmpos:PressurizedChamberPressure ;
  sosa:hasFeatureOfInterest  :wafer_373446766 ;
  sosa:madeOnPlatform        :machine_2 ;
```

```

cmpto:duringStep      :step_A_373446766 ;
sosa:hasSimpleResult  "67.14"^^xsd:float ;
cmpto:rawTimestamp    "481637082"^^xsd:decimal .

```

Key terms: cmpto:CMPObservation  $\sqsubseteq$  sosa:Observation · observable properties  $\sqsubseteq$  sosa:ObservableProperty (hierarchy in Section 4) · wafers, lots, wafer states  $\sqsubseteq$  sosa:FeatureOfInterest · tools and components  $\sqsubseteq$  sosa:Platform (via cmpto:Equipment) · recipes  $\sqsubseteq$  sosa:Procedure. cmpto:rawTimestamp keeps the source epoch value losslessly; sosa:resultTime may carry a derived xsd:dateTime. cmpto:machineDataId records the PHM MACHINE\_DATA acquisition-unit value.

## 4 Observable Property Hierarchy

Root: cmpto:CMPPParameter  $\sqsubseteq$  sosa:ObservableProperty. Groups, key leaves, and PHM grounding:

| Group                           | Leaf classes (selection)                                                                                                                           | PHM columns (altLabels)                                                                                       | Unit      |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-----------|
| Pressure                        | RetainingRing-,<br>MainOuterAirBag-,<br>CenterAirBag-, RippleAirBag-,<br>EdgeAirBag-,<br>PressurizedChamberPressure;<br>ContactPressure, Downforce | RETAINER_RING_PRESSURE,<br>MAIN_OUTER_/CENTER_/RIPPLE_/EDGE_AIR_BAG_PRESSURE,<br>PRESSURIZED_CHAMBER_PRESSURE | psi / N   |
| Velocity                        | PlatenRotationSpeed,<br>CarrierRotationSpeed,<br>WaferRotationSpeed,<br>RelativeVelocity                                                           | STAGE_ROTATION, HEAD_ROTATION, WAFER_ROTATION                                                                 | rpm / m/s |
| Slurry flow                     | SlurryFlowLineA/B/C                                                                                                                                | SLURRY_FLOW_LINE_A/B/C                                                                                        | ml/min    |
| Conditioning                    | DresserUsage,<br>DressingWaterStatus,<br>ConditioningSweepRate, -<br>Force, -Ratio                                                                 | USAGE_OF_DRESSER, DRESSING_WATER_STATUS                                                                       | misc.     |
| Usage<br>(cumulative)           | BackingFilm-, PolishingTable-,<br>DresserTable-, Membrane-,<br>PressurizedSheetUsage                                                               | USAGE_OF_* columns                                                                                            | count/min |
| Material<br>removal             | MaterialRemovalRate,<br>AverageRemovalRate                                                                                                         | AVG_REMOVAL_RATE                                                                                              | nm/min    |
| Temperature<br>/<br>environment | PadTemperature,<br>CleanroomTemperature,<br>CleanroomHumidity                                                                                      | — (synthetic completion)                                                                                      | °C / %    |
| Slurry<br>chemistry             | SlurryPH, -Viscosity,<br>AbrasiveParticleSize,<br>SolidContent,<br>ParticleConcentration,<br>IonicStrength, ZetaPotential                          | — (synthetic completion)                                                                                      | misc.     |
| Wafer<br>quality                | FilmThickness,<br>SurfaceRoughness,<br>WithinWaferNonUniformity,<br>WaferToWaferUniformity                                                         | — (synthetic completion)                                                                                      | nm / %    |

## 5 Entity Branches (Non-Observation)

- **Features of interest:** Wafer (Dummy-, Control-, StructuredWafer), Lot + LotType, WaferState (defects, actual removal), Layer/Material, notch/orientation attributes.

- **Platforms:** Equipment → Tool (incl. MetrologyTool: FTR, AFM, Ellipsometer, Profilometer, SEM) and ToolComponent (Platen, Carrier + six PressureZones, Chamber, Conditioner, ConditionerTable, SlurryPipe); CMPToolType (rotary/linear/orbital).
- **Consumables:** PolishingPad (padType, hardness, compressibility, lifetimes, GroovePattern), ConditionerDisc, RetainingRing, Membrane, PressurizedSheet, BackingFilm, Slurry (chemistry, AbrasiveType, ChemicalAdditive), DiWater, CleaningAgent.
- **Process:** Service, PolishingStep (stageLabel, stepIndex, hasNextStep, runsOnPlaten, timings), PolishingMode (BulkRemoval, FinePolishing, Buffing, Clearing), Recipe + SetPoint, EndPointDetection (optical/impedance/motor-current), PostCMPCleaning.
- **Quality & control:** ProcessCard (targets), ProcessWindow, SPCLimit, RunToRunControl, PrestonModel, MaterialSelectivity, ToolQualification, UsageTracking, DefectType hierarchy (Scratch, Dishing, Erosion, Corrosion, Over-/Underpolish).
- **Agents:** Agent + Role (ProjectOwner, ProcessOwner).

## 6 BFO Alignment Module (Optional)

cm-po-bfo-alignment-v2.ttl (~30 triples) maps core classes to BFO categories (object, process, role, quality, information content entity) and key properties to RO relations. It is NOT loaded into application graphs; its purpose is cross-ontology comparison — in particular with an independently developed, interview-based BFO-aligned CMP schema — without contaminating the SOSA-only core.

## 7 Dataset Coverage and Synthetic Completion

PHM 2016 populates: 20 observable-property leaves (verified 1:1 against all 28 CSV headers via the accompanying mapping table `csv_to_cm-po_v2.0_mapping.csv`), plus Wafer, PolishingStep (stages A/B), Tool, Chamber. Schema elements PHM does not populate (temperature/environment, slurry chemistry, wafer-quality metrics, and the entity branches Lot, Recipe/SetPoint, Slurry details, Pad details, Defects, EPD, ProcessCard, ProcessWindow/SPC, etc.) are completed with schema-conformant synthetic data (accompanying table `CMPO_v2.0_Synthetic_Completion_Table.xlsx`). Rule: synthetic records always carry explicit provenance labeling and are reported separately in any coverage or evaluation metric; they exercise the schema and pipeline and support no claims about physical CMP behaviour.

## 8 Versioning, Freeze and Gap Process

- **v2.0 is frozen** for the duration of the two Master's theses. Users do not modify the ontology.
- **Gap process:** missing or problematic elements are reported to `cm-po-gaps.md` in the shared repository (description, workaround, date). Gap reports are analysed as evidence about the construction method.
- **Patches:** only a defect that blocks work triggers v2.0.1 (changelog; delivered to all users on the same day). Content evolution is deferred to v3.
- **Research edition:** a BFO-integrated edition (`cm-po-v3.0-research.ttl`) exists for ontology-engineering research and cross-ontology comparison: identical content plus a BFO-2020 anchoring layer and IOF Core cross-references. Application projects and the Master's theses use v2.0; v3.0-research is not part of the frozen contract.

## 9 Disambiguation and Related Resources

- **Acronym disambiguation:** an unrelated ontology named CMPO exists — the "Configuration Management Process Ontology" of the SEON network (NEMO/UFES), covering software configuration management. This specification's CMPO (Chemical-Mechanical Planarization Process Ontology) is unrelated in domain and content; the full expansion is used on first mention in all documents.

- **Novelty status (checked July 2026):** no ontology for CMP/planarization process data was found in web and literature searches. Closest related resources: semantic modeling for semiconductor wafer fabrication (2023), ontologies for PHM of production systems (survey, JIM 2024), and a SOSA-based sensor ontology for forming processes (2024) — the latter confirming the "SOSA backbone + thin domain extension" pattern in another manufacturing domain. A registry check (LOV, BioPortal, IOF) is scheduled before external publication.
- **Manufacturing mid-level:** the Industrial Ontologies Foundry (IOF) Core (NIST, BFO-2020-based) is the community reference for manufacturing ontologies; the research edition of CMPO (Section 10) carries `rdfs:seeAlso` cross-references to IOF terms.

## 10 Known Limitations

- The physical semantics of the PHM MACHINE\_DATA column (`cmpos:machineDataId`, values 1–6) are dataset-specific and not fully documented by the data provider.
- `rdfs:comment` texts were AI-drafted and expert-reviewed; individual domain formulations may still be refined in v3.
- Sensor individuals are not modeled (PHM names no sensors); observations link to platforms directly via `sosa:madeOnPlatform`.
- The ontology targets batch-oriented historical data; streaming/windowing constructs are out of scope.
- Fab-workflow branches (ProcessCard, roles) reflect one fab's conventions and may not generalize.